Lost in Fog: Sensor Perturbations Expose Reasoning Fragility in Driving VLAs

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Abstract

Interpretable autonomous driving planners depend not only on generating explanations, but also on those explanations remaining reliable under real-world sensor degradation. In this paper we present a controlled perturbation study of Vision-Language-Action (VLA) robustness in autonomous driving, evaluating Alpamayo R1 (10B parameters) across 1,996 scenarios under eight sensor perturbations (Gaussian noise at four intensities, two lighting extremes, and two fog levels; $\sim$18,000 inference trials). We find that reasoning consistency is a high-fidelity indicator of trajectory reliability: when Chain-of-Causation (CoC) explanations change after perturbation, trajectory deviation spikes $5.3\times$ (21.8 m vs. 4.1 m), with $r = 0.99$ across attack types and $r_{pb} = 0.53$ per-sample (Cohen's $d = 1.12$). A controlled ablation provides evidence that enabling CoC generation is associated with improved trajectory accuracy (11.8% on average across conditions; $p < 0.0001$) under matched inference settings. Over the tested noise range ($\sigma \in \{10, 30, 50, 70\}$), degradation is approximately linear ($R^2 = 0.957$), while standard input preprocessing defenses provide only marginal relief. Together, these results establish CoC consistency as a quantitative proxy for planning safety and motivate reasoning-based runtime monitoring for safer VLA deployment.

1. Introduction

Vision-Language-Action models offer a capability that purely reactive planners cannot match: they explain their decisions. A VLA like Alpamayo R1 [28], the current state of the art for CoC-equipped driving models, does not merely predict a trajectory but it also states why: "Slow down because the lead vehicle is braking ahead in the same lane." For engineers certifying autonomous driving software under ISO 21448 (SOTIF), ISO PAS 8800 (safety of AI-based systems in road vehicles) [13], or NHTSA guidelines, this transparency matters. A model that articulates its reasoning can be interrogated, audited, and trusted in ways a black-box planner cannot.

Transparent reasoning offers little practical utility if the underlying model lacks robustness to sensor degradation. For example, consider a VLA deployed where morning fog degrades the camera feed. If the model still tracks the lead vehicle correctly, the explanation should say so. If it loses that cue and silently pivots to "Keep lane since the road ahead is clear," the trajectory will follow and the change may go undetected by any external monitor. This is precisely the failure mode that SOTIF analysis is designed to surface: a system behaving correctly in nominal conditions but degrading silently when inputs drift outside the operational design domain (ODD) (e.g. driving in fog is not considered as part of the ODD).

Adversarial robustness research in autonomous driving has mostly focused on perception components: traffic sign attacks [8], LiDAR spoofing [6], and camera patches [26] measure detection accuracy, not trajectory outcomes. VLA robustness studies [11, 17] target manipulation, not driving. Recent benchmarks [15] introduced sensor and prompt corruptions for driving VLMs, and ADvLM [31] showed gradient-based attacks can mislead them, but neither traces the causal chain from visual corruption through reasoning to trajectory error. That chain is what we measure.

A systematic analysis of this pipeline yields the four contributions of our paper:

1. **Chain-of-causation consistency as a trajectory reliability signal**: when perturbation changes the model's explanation, deviation spikes $5.3\times$ (21.8 m *vs.* 4.1 m), a relationship that holds both aggregate ($r = 0.99$, $n = 8$ attack types) and per-sample ($r_{pb} = 0.53$, point-biserial correlation, $n = 15{,}968$). To our knowledge, no prior work has shown that a VLA's explanations can flag corrupted predictions.
2. **A dose-response curve linking sensor noise to trajectory error** ($R^2 = 0.957$): ADE rises at 0.0048 m per unit σ, giving developers a direct formula for deployment risk assessment and operational design domain planning.
3. **Evidence for benefit of CoC generation**: a controlled ablation (same checkpoint, matched decoding settings)

shows that enabling CoC generation is consistently associated with lower ADE across all tested conditions (average 11.8%; $p < 0.0001$), with larger gains on complex maneuvers.

4. **Standard input sanitization via preprocessing defenses offers limited protection**: six off-the-shelf input-processing methods yield only marginal, statistically insignificant relief after Bonferroni correction, motivating VLA-specific defense research.

2. Related work

Vision-Language-Action Models for Driving. VLAs combine large vision-language models with robotic control to produce explainable driving systems. Earlier end-to-end driving networks [2] demonstrated that neural networks can steer directly from camera pixels, but offered little interpretability. RT-2 [4] showed that web-pretrained VLMs can transfer to robot actions. Octo [10] generalized this paradigm to multi-task settings. DriveLM [25] extended it to driving with graph-structured visual question answering. Alpamayo R1 [28] advanced the field with a 10B-parameter architecture generating chain-of-causation explanations alongside 64-waypoint trajectories at 10 Hz, reporting a 12% planning improvement in challenging closed-loop scenarios. Whether that reasoning holds under sensor degradation remains an open question.

Concurrent VLA Robustness Work. Concurrent work has begun to probe VLA robustness in adjacent domains. In manipulation, Guo *et al*. [11] test 17 perturbation types across four modalities, identifying the action head as most vulnerable; Eva-VLA [17] studies physical variations (table height, lighting angle) and reports failure rates exceeding 60%. In driving, RoboDriveVLM [15] benchmarks VLM trajectory prediction under six sensor corruptions and five prompt corruptions on nuScenes [5], finding fragility to prompt attacks. However, these studies largely focus on task-level outcomes; how explanation stability relates to trajectory error is outside their scope.

Adversarial Robustness in Perception. Existing robustness evaluations focus heavily on perception components, targeting modalities ranging from traffic signs [8], LiDAR [6], cameras [26] to sensor fusion [12]. A critical limitation of these approaches is their reliance on proxy metrics (e.g., detection AP, segmentation IoU) rather than system-level planning outcomes. Although recent work has begun to probe trajectory models either by perturbing input waypoints [30] or by applying gradient-based attacks on VLMs [31], the causal chain from natural sensor degradation, through reasoning instability, to downstream reasoning failure remains unexplored. We address this by quantifying the propagation of error from visual input, through the reasoning bottleneck, to the final trajectory.

Weather Robustness and Safety Evaluation. Determining the operational design domain of autonomous systems requires rigorous stress-testing against environmental corruption. Prior work has established benchmarks for foggy scene understanding [21, 22], enabled closed-loop weather simulation via CARLA [7], developed corruption-robustness protocols [18], and contributed datasets such as KITTI-360 [16] and augmentation strategies [27]. However, these contributions are predominantly evaluated on perception-level metrics (e.g., semantic segmentation mIoU or detection AP), rather than on planning outcomes. Thus, they do not quantify how such environmental corruptions propagate through the complex reasoning chains of Vision-Language-Action models to impact downstream trajectory fidelity.

Explainability as a Safety Mechanism. While the importance of explainable planning is widely recognized [1], traditional approaches rely on post-hoc mechanisms (e.g., attention visualization [14] or saliency maps [3]) that approximate reasoning rather than exposing it. VLAs fundamentally alter this paradigm by generating intrinsic Chain-of-Causation explanations directly alongside control outputs. An unresolved question, however, is whether this reasoning channel remains robust to perturbations. We address this gap by establishing CoC consistency as a quantitative proxy for trajectory fidelity (Sec. 4).

3. Experimental setup

We evaluate VLA robustness by applying controlled sensor corruptions to real driving sequences and measuring the resulting degradation in both trajectory accuracy and reasoning consistency. The evaluation spans 1,996 scenarios across eight perturbation conditions, and includes a CoC-ablation study, totaling approximately 18,000 inference trials for the primary evaluation.

3.1. Model and dataset

Model. For our experiments we adopt Alpamayo R1 [28], a 10B-parameter VLA architecture grounded on the Qwen3-VL vision encoder. The model processes multi-view camera feeds via dynamic resolution encoding (164K-197K pixels, aspect-ratio adaptive), fuses visual context with ego-history to jointly predict a 64-waypoint trajectory at 10 Hz ($T = 6.4$ s) and a corresponding chain-of-causation explanation.

Dataset. The evaluation is performed on the PhysicalAI-Autonomous-Vehicles validation split. We uniformly sample from the 1,996 driving sequences spanning diverse maneuvers, including lane-keeping, vehicle-following, and unprotected turns. Sample sizes range from $n = 475$

Table 1. Eight perturbation conditions spanning three corruption modalities, applied synchronously across all camera views. Categories align with the additive, multiplicative, and physics-based corruption types described in Sec. 3.2.

Category	Name	Implementation	Real-World Correlation
Sensor Noise	noise_10	Gaussian $\sigma = 10$	Typical CMOS noise
	noise_30	Gaussian $\sigma = 30$	Moderate degradation
	noise_50	Gaussian $\sigma = 50$	Heavy sensor noise
	noise_70	Gaussian $\sigma = 70$	Severe degradation
Photometric Shift	dark	Brightness $0.4\times$	Tunnel entry, dusk
	bright	Brightness $1.6\times$	Sun glare
Volumetric Scattering	fog_light	Scattering $\alpha = 0.3$	Light haze
	fog_heavy	Scattering $\alpha = 0.7$	Dense fog

(FOLLOW_VEHICLE) to $n = 14$ (TURN_LEFT); to ensure statistical power, our per-category analysis is restricted to categories with $n \geq 100$.

Baseline. We compare against a constant-velocity physics baseline that propagates each clip's recorded ego velocity as a straight-line trajectory over the 6.4-second horizon, producing 64 uniformly-spaced waypoints with no steering or acceleration adjustment. This baseline serves to distinguish learned semantic planning from trivial kinematic continuation; the 68.3% improvement it yields should therefore be read as the gap over kinematic extrapolation, not over learned planners. A more informative comparison against learned trajectory predictors such as AD-MLP [29] is a natural extension for future work.

Scenario taxonomy. To enable granular robustness analysis, we stratify the dataset into distinct behavioral categories based on semantic action primitives extracted from the ground-truth (clean) CoC explanations. The resulting distribution shows common urban interactions: Vehicle Following ($n = 475$), Intersection Navigation ($n = 344$), Signal Compliance ($n = 302$), and Lane Keeping ($n = 213$). Complex maneuvers such as Passing ($n = 177$) and Turns (Right/Left, combined $n = 54$) are also represented, with the remaining tail encapsulated in an Other category ($n = 418$).

3.2. Perturbation design

We evaluate robustness across an eight-condition threat model spanning three corruption modalities (Tab. 1). To preserve geometric consistency across the VLA's multi-view input, we apply each perturbation synchronously across all camera views.

Sensor Noise (Additive). We inject Gaussian noise ranging from $\sigma = 10$ (simulating typical CMOS read noise [9]) to $\sigma = 70$ (simulating severe low-light gain amplification).

Photometric Shift (Multiplicative). We scale global image intensity linearly ($0.4\times$, $1.6\times$) to simulate rapid exposure changes typical of tunnel entry and direct sun glare, high-risk scenarios empirically linked to intersection crashes [19].

Volumetric Scattering (Physics-based). We synthesize fog using the Narasimhan and Nayar atmospheric scattering model [20]. We calibrate the scattering coefficient α to meteorological standards: $\alpha = 0.3$ approximates light haze ($\sim$500 m visibility), while $\alpha = 0.7$ approximates dense fog ($\sim$100 m visibility).

3.3. Evaluation metrics

We quantify system robustness using four complementary metrics that capture both task performance and reasoning stability:

1. **ADE (Average Displacement Error):** Mean L2 distance between predicted and ground-truth waypoints across the 64-step horizon:

$$\text{ADE} = \frac{1}{T} \sum_{t=1}^{T} \|\hat{\mathbf{p}}_t - \mathbf{p}_t\|_2 \tag{1}$$

2. **ADE Degradation** (ΔADE): Increase relative to clean-condition prediction.
3. **CoC Change Rate:** Fraction of samples whose natural-language explanation differs after perturbation. We use exact string matching after whitespace normalization; even a single changed word counts.
4. **L2 Trajectory Deviation:** L2 norm between clean-condition and perturbed predicted trajectories: $\|\hat{\mathbf{T}}_{\text{clean}} - \hat{\mathbf{T}}_{\text{attack}}\|_2$.

We prioritize ADE over Final Displacement Error (FDE) to capture deviation dynamics throughout the entire 6.4-second planning window, though FDE trends remain consistent with ADE across all trials.

Binary *vs*. Continuous CoC Matching. We evaluated three approaches for quantifying explanation change (binary string matching, sentence-embedding cosine similarity (all-MiniLM-L6-v2), and Jaccard word overlap) across all 15,968 sample-attack pairs. Sentence-embedding similarity achieves Spearman $\rho = -0.54$ against L2 trajectory deviation, comparable to binary point-biserial correlation ($r_{pb} = 0.53$); Jaccard achieves $|r| = 0.45$. We adopt binary matching for two reasons. This distinction is safety-relevant: for short CoC explanations (7-30 tokens), a single action-verb change can invert driving intent while still yielding high embedding similarity (e.g. switching from "keep distance to the lead vehicle" to "keep lane since the road is clear" yields cosine similarity above 0.85 but reflects fundamentally different driving intent). Binary matching reliably flags these high-risk semantic inversions in cases where continuous metrics signal false stability.

3.4. Reproducibility and compute

Threat Model & Scope. This study restricts perturbations to natural sensor degradations: noise, fog, and lighting extremes, rather than gradient-based adversarial attacks. This scope reflects three considerations: (1) environmental corruptions such as rain, fog, and sun glare occur in routine driving, whereas gradient-based attacks (FGSM, PGD) remain largely confined to academic settings; (2) ISO 21448 (SOTIF) and NHTSA guidelines mandate robustness to operational design domain conditions, making these perturbations directly relevant to certification requirements; (3) deterministic perturbation functions ensure reproducible benchmarking. Black-box and gradient-based attacks, shown effective against driving VLMs by Zhang *et al*. [31], are deferred to future work.

Ablation Protocol. To disentangle correlation from causation, we conduct a controlled ablation in which every sample is re-evaluated in a trajectory-only inference mode (CoC generation suppressed). We use the same model checkpoint, identical decoding hyperparameters (temperature$=0.6$, top-$p=0.98$, seed$=42$, fixed via TORCH CUDA manual seed), and a fixed random seed to evaluate the effect of explicit reasoning on planning fidelity. Because the seed is reset identically before clean-condition and perturbed-condition inference, each comparison is fully deterministic per sample; run-to-run sampling variance does not contribute to the measured CoC change rate.

Ablation Validity. The goal of the CoC ablation is to isolate the effect of generating an explicit natural-language chain-of-causation from other inference-time factors. To this end, both "With CoC" and "Without CoC" runs use the same model checkpoint, identical visual inputs and perturbations, and identical decoding hyperparameters, with a fixed random seed to minimize sampling variance. The only change between conditions is the token generation budget: the "With CoC" condition generates up to 512 tokens, allowing full chain-of-causation output; the "Without CoC" condition is limited to 1 token, suppressing all CoC text while leaving model weights, visual inputs, and decoding hyperparameters unchanged. We acknowledge that this constitutes a token-budget change rather than a pure reasoning toggle, and report these findings as evidence under this specific protocol, while acknowledging that stronger causal identification would require additional controls (e.g., prompt-level suppression or token-matched dummy generation) beyond the scope of this paper.

Computational Infrastructure. All inference was executed on a single NVIDIA A100-40GB GPU. Wall-clock latency per sample ranges from 8-15 s, assuming resident model memory and pre-computed perturbations. The total computational budget is approximately 60 GPU-hours: 40 hours for the primary evaluation and 20 hours for the ablation study. The reduced runtime for the latter case reflects the lower token generation overhead of the trajectory-only inference mode.

Table 2. The VLA reduces ADE by 68.3% relative to a constant-velocity baseline on clean inputs ($n=1{,}996$; $p < 10^{-257}$), establishing a strong performance floor before any corruption is introduced.

Model	ADE (m)	Std	p-value
Physics (const. velocity)	6.32	4.87	-
Alpamayo R1 (VLA)	2.00	1.89	-
Improvement	**68.3%**	-	$< 10^{-257}$

Table 3. Noise $\sigma=70$ is the dominant threat (ΔADE$=+0.30$ m, 52.7% CoC flips); mild noise ($\sigma=10$) is statistically negligible ($p=0.637$). Eight conditions ranked by ΔADE, $n=1{,}996$ each. Cohen's d against the clean baseline ranges 0.005-0.149; the >5 m column is the fraction of clips where $\|\hat{\mathbf{T}}_{\text{clean}} - \hat{\mathbf{T}}_{\text{attack}}\|_2 > 5$ m, capturing tail risk more faithfully than mean ADE. $^{\dagger}p > 0.05$ before correction.

Attack	ADE	ΔADE	CoC Δ	$>$5m	p
Noise $\sigma=10$	2.01	+0.01	18.1%	18.8%	0.637
Dark ($0.4\times$)†	2.06	+0.05	39.0%	51.7%	0.058
Light Fog ($\alpha=0.3$)	2.06	+0.06	16.9%	18.1%	0.006
Bright ($1.6\times$)	2.06	+0.06	31.9%	34.7%	0.025
Noise $\sigma=30$	2.07	+0.07	35.0%	48.5%	0.020
Heavy Fog ($\alpha=0.7$)	2.09	+0.09	33.8%	47.3%	0.001
Noise $\sigma=50$	2.16	+0.15	45.4%	61.0%	4×10^{-7}
Noise $\sigma=70$	2.30	+0.30	52.7%	70.6%	2×10^{-17}

4. Results

Under clean conditions, Alpamayo R1 achieves 2.00 m ADE ($\sigma=1.89$) compared to 6.32 m ($\sigma=4.87$) for the constant-velocity baseline, a 68.3% improvement (Tab. 2; paired $t=36.97$, $p < 10^{-257}$). This places the VLA in a fundamentally different performance regime from naive kinematics, retaining the majority of this advantage even under severe perturbation.

4.1. Perturbation robustness

Attack Impact. Table 3 ranks all eight perturbations by trajectory degradation. Heavy noise ($\sigma = 70$) is the clear outlier: +0.30 m ADE, 52.7% CoC change rate, and 70.6% of samples exceeding 5 m L2 deviation. At the other end, mild noise ($\sigma = 10$) is essentially invisible to the model (+0.01 m, not significant).

The split between CoC change rate and trajectory impact is not uniform: lighting and fog perturbations cause disproportionately high CoC change rates relative to their trajectory impact. Dark conditions flip 39% of explanations but only degrade ADE by 0.05 m. This decoupling (the model rephrases its reasoning yet still lands near the right trajectory) is further characterized in the CoC-Trajectory Correlation paragraph below.

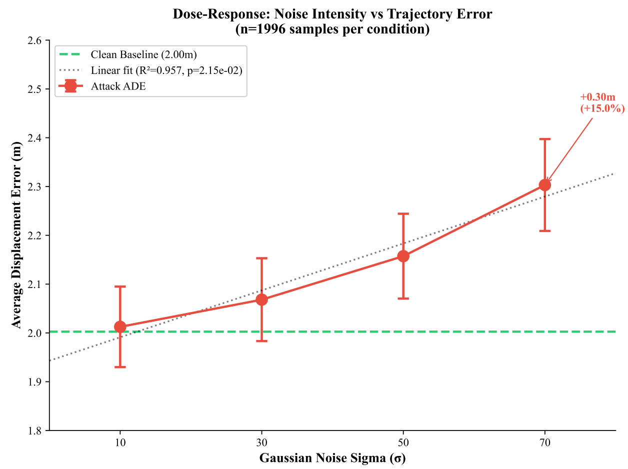

Figure 1. ADE grows linearly with noise intensity ($R^2 = 0.957$, slope $= 0.0048$ m/σ; four noise levels), enabling a direct risk-assessment formula: each 10-unit σ increment costs ≈ 0.05 m ADE. Error bars: 95% CI.

Table 4. ADE rises linearly with noise intensity ($R^2 = 0.957$), adding approximately 0.05 m per 10-unit σ increment. Values are mean ADE with 95% bootstrap confidence intervals ($n = 1{,}996$ per level).

$\boldsymbol{\sigma}$	n	**ADE (m)**	**ΔADE**	**95% CI**
10	1,996	2.01	+0.01	[1.93, 2.10]
30	1,996	2.07	+0.07	[1.98, 2.15]
50	1,996	2.16	+0.15	[2.07, 2.24]
70	1,996	2.30	+0.30	[2.21, 2.40]

Dose-Response. The four noise levels form a natural dose-response curve (Fig. 1). A linear regression on σ *vs.* mean ADE yields $R^2 = 0.957$ ($p = 0.022$), with slope 0.0048 m per unit σ. Each 10-unit σ increment adds roughly 0.05 m to trajectory error. While moderate ADE increases could partially reflect appropriate conservatism under degraded inputs, the monotonic escalation of the >5 m tail from 18.8% to 70.6% across noise levels captures large, unpredictable prediction shifts inconsistent with systematic safety behavior.

Four data points is a thin basis for asserting linearity. Formal model comparison (AIC) on the four data points favors the linear fit over log-linear (ΔAIC $= +6.3$), power-law (ΔAIC $= +6.1$), and saturating (ΔAIC $= +8.8$) alternatives, suggesting linearity is adequate within the tested range, though additional noise levels ($\sigma = 20, 40, 60, 80+$) would firm this up considerably. That said, the practical takeaway stands: degradation is monotonic and the slope gives a useful rule of thumb for deployment risk assessment.

CoC-Trajectory Correlation. Partitioning all 15,968 attacked samples (8 attacks $\times$ 1,996 clips) by whether the CoC explanation changed or stayed identical after perturbation produces the central result of this study.

When the model's explanation stays put, the trajectory barely budges: 4.13 m mean (2.16 m median) deviation. When the explanation flips, deviation jumps to 21.82 m mean (15.39 m median): a 5.3$\times$ mean ratio and 7.1$\times$ median ratio, both robust to the right-skewed deviation distribution ($t = -78.1$, Cohen's $d = 1.12$). Across attack types, the correlation is $r = 0.99$ ($n = 8$), indicating that perturbations that induce more CoC changes produce proportionally larger trajectory errors with near-perfect linearity (Fig. 2). We treat CoC flips as a binary alarm runtime monitoring signal and evaluate its detection characteristics in our open-loop, paired clean-vs-perturbed setting (Tab. 7).

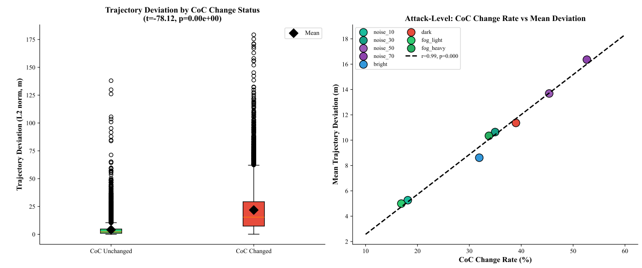

Figure 2. CoC explanation stability is strongly associated with trajectory integrity: preserved explanations yield 4.1 m mean deviation, while changed explanations yield 21.8 m (5.3$\times$; $r_{pb} = 0.53$, $n = 15{,}968$, left). The relationship holds across attack types ($r = 0.99$, $n = 8$, right). Detection characteristics of this signal are reported in Tab. 7.

Table 5. When a perturbation flips the VLA's CoC explanation, mean trajectory deviation is 5.3$\times$ higher than when the explanation is preserved (21.82 m *vs.* 4.13 m mean; 7.1$\times$ by median); Cohen's $d = 1.12$, $p < 10^{-100}$, $n = 15{,}968$.

Condition	**Mean L2**	**Median L2**	$\boldsymbol{n}$
CoC Unchanged	4.13 m	2.16 m	10,525
CoC Changed	21.82 m	15.39 m	5,443
Ratio	5.3$\times$	7.1$\times$	-
t-test / p	$t = -78.1$,	$p < 10^{-100}$	-
Cohen's d	1.12		-

Table 6. Per-attack point-biserial correlation between CoC stability and L2 trajectory deviation ($n = 1{,}996$ per attack; all $p < 10^{-70}$). Mild perturbations yield the strongest per-sample diagnostic signal ($r_{pb} > 0.63$, Ratio $> 12\times$). Under severe noise ($\sigma = 70$), even unchanged CoC explanations co-occur with elevated baseline deviation (8.6 m), compressing the diagnostic gap to 2.7$\times$.

Attack	$\boldsymbol{r_{pb}}$	**Changed/Unchanged**
Light Fog ($\alpha = 0.3$)	0.643	13.5$\times$
Noise $\sigma = 10$	0.639	12.8$\times$
Bright (1.6$\times$)	0.564	7.5$\times$
Heavy Fog ($\alpha = 0.7$)	0.504	4.5$\times$
Noise $\sigma = 30$	0.501	4.6$\times$
Dark (0.4$\times$)	0.481	4.1$\times$
Noise $\sigma = 50$	0.438	3.2$\times$
Noise $\sigma = 70$	0.384	2.7$\times$

Table 7. CoC flip as an alarm signal across all eight conditions (alarm = CoC changed; unsafe event = L2 deviation > 5 m, $n =$ 1,996 per attack). Precision remains ≥ 0.79 throughout; recall and AUROC degrade under severe noise as language and trajectory branches decouple.

Attack	Prec	Recall	FPR	AUROC
Noise $\sigma = 10$	0.818	0.787	0.041	0.873
Noise $\sigma = 30$	0.834	0.601	0.113	0.744
Noise $\sigma = 50$	0.838	0.624	0.189	0.717
Noise $\sigma = 70$	0.849	0.633	0.271	0.681
Dark ($0.4\times$)	0.842	0.635	0.128	0.754
Bright ($1.6\times$)	0.791	0.728	0.102	0.813
Light Fog ($\alpha = 0.3$)	0.819	0.765	0.037	0.864
Heavy Fog ($\alpha = 0.7$)	0.841	0.600	0.102	0.749
Aggregate	**0.832**	**0.647**	**0.102**	**0.773**

CoC as a Runtime Monitor. Runtime safety monitors are well-studied external mechanisms that operate at inference time alongside a deployed model, flagging potential safety concerns without modifying the underlying model weights [24]. We evaluate CoC consistency as a binary runtime monitor alarm: an alert is raised whenever the Chain-of-Causation (CoC) explanation changes after perturbation, and an event is flagged as unsafe if $\|\hat{\mathbf{T}}_{\text{clean}} - \hat{\mathbf{T}}_{\text{perturbed}}\|_2 > 5$ m (matching the tail-risk threshold in Tab. 3).

Note: This "monitor" is evaluated in an open-loop paired clean-vs-perturbed protocol; we do not claim an online deployable monitor in the current setup. Table 7 reports precision, recall, false positive rate (FPR), and area under the receiver operating characteristic curve (AUROC), computed from the same per-sample binary outcomes; AUROC uses per-sample word-similarity as a graded score. Under mild perturbations (the operationally realistic regime for early warning), the binary alarm shows strong performance: Noise $\sigma = 10$ achieves AUROC$= 0.873$ with FPR$= 0.041$, capturing 78.7% of unsafe events while raising a false alarm on fewer than 1-in-24 nominal samples. Under severe noise ($\sigma = 70$), recall falls to 63.3% and FPR rises to 0.271, reflecting a language-trajectory decoupling in which the trajectory decoder fails via collapsing kinematic priors while the language branch continues producing coherent but safety-irrelevant explanations. We hypothesize this occurs because the language decoder relies on high-probability text priors that dominate weakened visual tokens, producing fluent but perceptually ungrounded outputs, while the trajectory decoder operates more directly on degraded visual features without such a fallback. Precision remains high (≥ 0.79) across all conditions: when the monitor does alarm, it is reliable; the primary safety limitation is recall under severe degradation.

Toward Deployable Monitoring. The paired clean-vs-perturbed protocol above is an evaluation construct; in deployment no uncorrupted reference frame is available. Two strategies can bridge this gap: *temporal consistency* (comparing CoC explanations across consecutive frames via semantic similarity, where an abrupt shift absent a corresponding scene change signals potential degradation) and a *learned surrogate* (a lightweight classifier on CoC token embeddings trained to predict elevated trajectory deviation directly, using the paired data in this study as supervision). Validating these strategies is an immediate priority for future work.

4.2. Failure analysis and scenario vulnerability

Qualitative Breakdown. Under severe noise ($\sigma = 70$), changed CoC explanations fall into three major categories: action-word changes (27.8%), object-reference changes (33.4%), and shifted object focus (22.2%). Across all eight attack types, 98.9% of flipped explanations involve at least one action-word or object-reference change, both categories with direct trajectory implications. Pure paraphrase flips (changed wording, preserved intent) represent a negligible fraction, confirming that exact-match sensitivity does not conflate harmless rephrasing with safety-relevant intent inversions. A representative failure illustrates the case: "Keep distance to the lead vehicle because it is directly ahead" becomes "Keep lane to continue driving since no critical agent needs attention." The model fails to track the lead vehicle and downgrades the scenario from vehicle-following to free cruising, omitting the necessary deceleration required for safe following.

Critically, perturbed explanations show no calibrated uncertainty: the model asserts phrases such as "since the lane ahead is clear" with essentially identical confidence under pristine and corrupted inputs. This absence of epistemic signaling represents a safety-critical gap since reasoning failures may appear linguistically well-justified even when the underlying perception has degraded.

A contrasting pattern under photometric shift is instructive. Dark conditions ($0.4\times$ brightness) flip 39% of CoC explanations, yet ADE degrades by only 0.05 m ($p = 0.058$, not significant). This dissociation suggests CoC instability can register distributional shift before the trajectory decoder is visibly affected: the reasoning channel pivots while ego-history and kinematic priors hold the trajectory near its clean-condition path. Table 7 confirms this: the Dark-condition monitor achieves AUROC$= 0.754$ with FPR$= 0.128$, detecting 63.5% of high-deviation events despite only a 0.05 m mean trajectory impact, confirming that CoC instability can flag photometric distribution shift before trajectory metrics visibly degrade.

Scenario Vulnerability. Figure 3 maps ADE degradation across all eight attack types and seven scenario categories.

STOP_SIGNAL and FOLLOW_VEHICLE show the steepest degradation, which is particularly worrisome given their

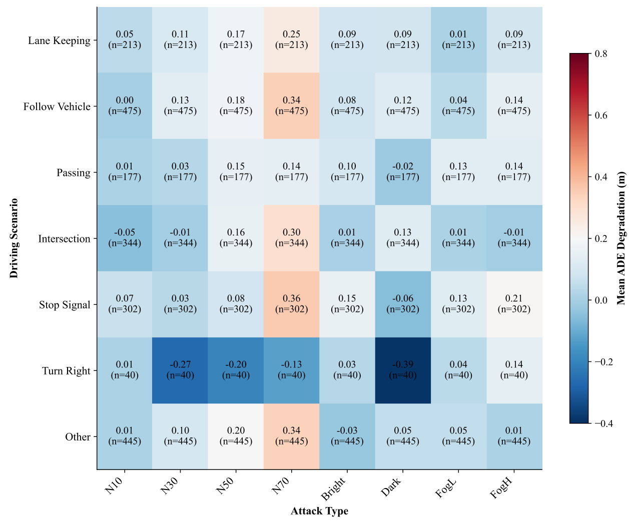

Figure 3. Safety-critical scenarios sustain the greatest degradation across all eight attack types: STOP_SIGNAL (ΔADE up to $+0.36$ m) and FOLLOW_VEHICLE ($+0.34$ m) are the consistently hottest rows regardless of whether the perturbation is noise, lighting, or fog. PASSING is deceptively resilient in trajectory terms ($+0.14$ m under heavy noise) yet carries the highest CoC change rate (58.2%) among high-n scenarios: another instance of reasoning being more fragile than the trajectory it guides. *TURN_RIGHT ($n=40$) shown for completeness; excluded from significance tests ($p=0.31$).

safety criticality. PASSING scenarios hold up reasonably well in trajectory terms, though their CoC change rate (58.2%) is the highest among scenarios with $n \geq 100$, another example of reasoning being more fragile than the trajectory itself.

A counterintuitive pattern reinforces this concern: FOLLOW_VEHICLE simultaneously holds the lowest clean-condition ADE (1.75 m, the best baseline performance) and the highest perturbation sensitivity ($+0.34$ m). This suggests the model relies most heavily on visual cues in scenarios where the task is well-defined. Tracking a lead vehicle provides a strong visual anchor under clean conditions but becomes the specific point of failure when sensor quality degrades. Scenarios where the VLA performs best are also the most brittle under distribution shift. Finally, a cross-attack consistency analysis reveals that 21.4% of clips fail under three or more attack types, indicating systematic architectural vulnerability rather than random sensitivity to individual perturbations.

4.3. Ablation and defense baselines

CoC Generation Benefit. Across all nine perturbations, enabling CoC generation reduces ADE (Tab. 8). The average improvement is 11.8% (range: 9.1-15.8%); effect sizes are small (mean $d_z = 0.14$) but consistent without exception. Lane-change maneuvers benefit most ($d_z = 0.43$, $\Delta\text{ADE} = 0.91$ m, $p < 0.001$), while simple lane-keeping benefits least, consistent with the intuition that explicit reasoning is helpful for complex decisions. CoC-enabled trajectories also degrade 15% more slowly under increasing noise (slope $0.0039\,\text{m}/\sigma$ *vs.* $0.0046\,\text{m}/\sigma$), suggesting reasoning anchors predictions against distribution shift.

Table 8. Enabling CoC generation is associated with reduced ADE by 11.8% on average, consistent across every condition including all eight attacks (mean Cohen's $d_z = 0.14$; all $p < 0.0001$, paired t-test with Wilcoxon confirmation).

Condition	With CoC	W/o CoC	ΔADE	d_z
Clean	2.01	2.31	+0.30	0.15
Noise $\sigma = 10$	2.02	2.25	+0.23	0.12
Noise $\sigma = 30$	1.97	2.34	+0.37	0.18
Noise $\sigma = 50$	2.10	2.42	+0.31	0.16
Noise $\sigma = 70$	2.23	2.53	+0.30	0.14
Dark (0.4×)	2.04	2.37	+0.32	0.16
Bright (1.6×)	2.03	2.26	+0.23	0.11
Light Fog	2.02	2.22	+0.20	0.11
Heavy Fog	2.09	2.34	+0.25	0.12
Average	**2.06**	**2.33**	**+0.28**	**0.14**

Preprocessing Defenses. We evaluated six off-the-shelf input-preprocessing defenses; Tab. 9 reports all six alongside the no-defense baseline.

Improvements range from 0.5-0.7 m, not statistically significant after Bonferroni correction.

Severity-Conditioned Effect. A conditional analysis reveals a consequential interaction: preprocessing defenses reduce L2 deviation by 9.8 m for samples with severe degradation (L2 $>$ 30 m) but increase it by 7.9 m for samples with mild degradation (L2 $<$ 10 m), likely due to smoothing-induced distribution shift. Since mild degradation is far more common in practice, a naive "always apply defense" policy is contraindicated and can increase operational risk. These results motivate adaptive, severity-conditioned defense strategies rather than uniform preprocessing.

Table 9. No off-the-shelf preprocessing defense achieves significance after Bonferroni correction; the best average improvement is 0.7 m. All six evaluated defenses are shown, ordered by average L2 improvement. Values are mean L2 deviation between clean and perturbed predicted trajectories (not ADE); lower = better.

Defense	N $\sigma = 30$	N $\sigma = 70$	Dark	Fog	Avg Δ
No Defense	21.6	23.7	21.9	21.1	-
Bilateral	20.9	22.7	20.9	21.2	+0.7
Gaussian 3×3	20.8	23.0	21.4	20.9	+0.6
Gaussian 5×5	20.9	23.3	21.3	20.9	+0.5
JPEG Q75	21.4	23.2	21.4	20.3	+0.5
Median 3×3	21.0	23.4	21.4	20.8	+0.4
Median 5×5	21.1	23.5	21.6	21.0	+0.3

5. Discussion

While the aggregate ADE degradation appears modest (15%, from 2.00 m to 2.30 m), mean-centric statistics fail to capture the severity of the tail distribution under sensor degradation: 70.6% of samples exceed 5 m L2 deviation under heavy noise, sufficient for multi-lane departure at highway speeds. In a safety-critical system, this kind of tail risk is often more consequential than the mean, because rare but large deviations can dominate operational hazard.

A related concern is that perturbed explanations provide no signal of input degradation. The model asserts statements such as "the lane ahead is clear" with essentially identical confidence whether the camera feed is pristine or corrupted, revealing a lack of calibrated epistemic signaling in the language channel. Therefore, uncertainty-aware language supervision, where models hedge under degraded inputs, represents a promising training direction. The language-trajectory decoupling observed under severe noise also defines the practical boundary of CoC-based monitoring: explanations serve as a reliable safety proxy in the mild-to-moderate perturbation regime, where the language branch still tracks perception, but become less reliable under extreme degradation where text priors dominate. The coupling between CoC stability and trajectory fidelity (Tab. 5) is correlational rather than causal: the language and trajectory branches may share upstream representations without one governing the other, a trustworthiness gap that certification frameworks such as ISO PAS 8800 [13] must address.

These findings suggest three data augmentation priorities for improving robustness: heavy Gaussian noise ($\sigma \geq 50$), low-light conditions (which trigger 39% CoC change rates despite modest trajectory impact), and targeted collection for STOP_SIGNAL and FOLLOW_VEHICLE scenarios.

5.1. Limitations

Methodological Constraints. The perturbations are synthetic and applied independently per frame; real sensor degradation carries temporal correlations, rolling-shutter artifacts, and hardware-specific noise characteristics that are not captured in our stimuli. Moreover, the evaluation is open-loop: each clip receives a corrupted initial frame with no subsequent feedback, so cascading errors are not captured.

Architectural Generalizability. While Alpamayo R1 serves as a representative state-of-the-art VLA, the transferability of these findings to alternative architectures (e.g., RT-2, PaLM-E) remains an open empirical question. Similarly, although the constant-velocity baseline effectively isolates learned reasoning from kinematic extrapolation, future work should benchmark against learned trajectory predictors (e.g., Trajectron++ [23]) to disentangle VLA-specific vulnerabilities from general forecasting errors.

Statistical & Computational. The noise dose-response analysis is based on four noise levels; adding intermediate intensities would strengthen or refute the linearity assumption ($R^2 = 0.957$). Similarly, the attack-level correlation ($r = 0.99$) is computed over only eight attack types; additional perturbation modalities would confirm whether near-perfect linearity generalizes to a broader threat model. VLA inference at 10B parameters requires 8-15 s per frame on an A100 GPU, far below real-time requirements; the robustness properties of quantized or distilled variants remain an open research question.

5.2. Future work

Building on the monitoring results in Tab. 7, promising directions include temporal CoC consistency checks, perturbation probing at inference time, and ensemble consensus over CoC outputs, as well as per-scenario breakdown of operating characteristics beyond the aggregate metrics reported here. Closed-loop evaluation in a simulation environment (e.g., CARLA, nuPlan) is a natural next step: it would enable direct collision and off-road rate computation (not derivable from our open-loop setup, where the >5 m deviation tail serves as a calibrated operational proxy) and capture cascading errors that accumulate when corrupted predictions feed back into subsequent frames.

6. Conclusion

In this paper we present a systematic stress-test of Vision-Language-Action (VLA) robustness in the context of autonomous planning. By evaluating the Alpamayo R1 architecture across a stratified set of 1,996 driving scenarios and eight distinct sensor corruptions (totaling ~18,000 primary inference trials), we quantify the specific vulnerabilities of end-to-end reasoning models under domain shift. Our analysis shows that standard input defenses yield no statistically significant improvements, while performance under Gaussian noise follows a predictable linear degradation profile ($R^2 = 0.957$), providing a calibrated heuristic for deployment risk. Our primary finding is that reasoning consistency serves as a high-fidelity proxy for planning safety. We observe a strong coupling between linguistic stability and trajectory error ($r = 0.99$ across modalities; $r_{pb} = 0.53$ per-sample), where reasoning collapse signals a $5.3\times$ spike in displacement error (21.8 m *vs.* 4.1 m; Cohen's $d = 1.12$). A controlled ablation provides evidence consistent with a benefit of CoC generation: trajectory accuracy is associated with an 11.8% improvement across all conditions ($p < 0.0001$). Together, these results establish CoC consistency as a quantitative proxy for trajectory fidelity and motivate both VLA-specific defense mechanisms and reasoning-based runtime monitoring as research directions for improving autonomous driving safety.

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